

Ridge regression

Penalise the size of θ : the L2 penalty and its budget form, the closed form $(X^T X + \lambda I)^{-1} X^T y$, shrinkage, invertibility, the intercept and choosing λ .

THE QUESTION

A high-degree polynomial fit chases the noise and its coefficients explode. How do we keep θ small and still fit the data?

THE INTUITION

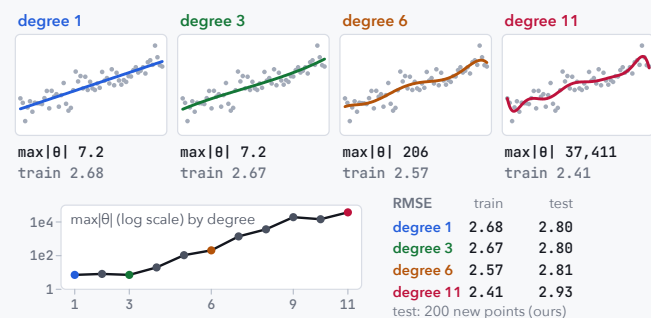
Add $\lambda \theta^T \theta$ to the squared error. Every unit of coefficient size now costs something, so the fit keeps only the size the data pays for. It is the same as minimising the error inside a ball $\theta^T \theta \leq S$, and the λ on the diagonal makes the matrix invertible.

KEY TAKEAWAY

$\hat{\theta} = (X^T X + \lambda I)^{-1} X^T y$. $\lambda = 0$ is least squares and $\lambda \rightarrow \infty$ gives 0; in between every coefficient shrinks, none to exactly 0. Leave θ_0 out of the penalty and pick λ by cross-validation.

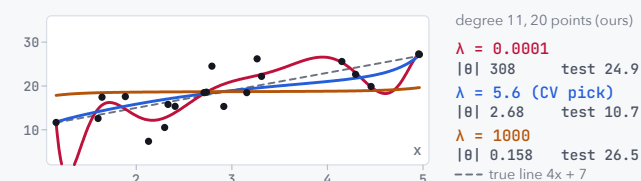
fit degree d $\rightarrow$ large $|\theta|$ overfits $\rightarrow$ add $\lambda \theta^T \theta$ $\rightarrow$ $(X^T X + \lambda I) \theta = X^T y$ $\rightarrow$ choose λ by CV

1 Higher degree: exploding coefficients



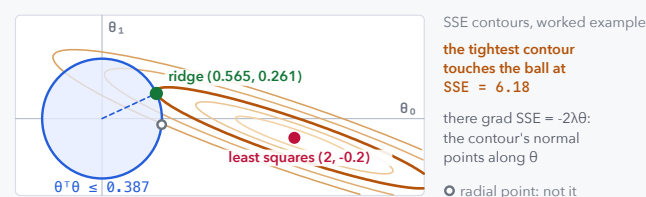
The ridge notebook's 60 points: $y = 4x + 7$ plus noise with $\sigma = 3$, x from 60° to 296° in radians. Least squares on the columns 1, x , ..., x^d . From degree 1 to 11 the training RMSE falls from 2.68 to 2.41 while $\max|\theta_j|$ climbs from 7.2 to 37,411 (the slides' chart). On 200 fresh points the test RMSE rises from 2.80 to 2.93. Training error well below test error is overfitting. Part of the growth is scale, since x^{11} reaches $7 \cdot 10^7$ here: standardise the columns before penalising them.

2 Ridge charges for the size of θ



$J(\theta) = (y - X\theta)^T (y - X\theta) + \lambda \theta^T \theta$, $\lambda \geq 0$, the SSE plus $\lambda \|\theta\|_2^2$. $\lambda = 0$ is plain least squares, a small λ shrinks θ a little, a large one a lot, and $\lambda \rightarrow \infty$ drives θ to 0. More λ means more bias and less variance. In the figure, $\lambda = 10^{-4}$ leaves $\|\theta\| = 308$ and wiggles (test MSE 24.9), $\lambda = 5.6$ gives 2.7 and 10.7, and $\lambda = 1000$ is nearly flat (26.5). The figure fits degree 11 on standardised powers with the intercept unpenalised (panel 9). The slides call the data term MSE; as written it is the sum of squares, which only rescales λ by N .

3 The same problem with a budget: $\theta^T \theta \leq S$



Minimise the SSE subject to $\theta^T \theta \leq S$. Its Lagrangian is $\text{SSE} + \lambda(\theta^T \theta - S)$ with $\lambda \geq 0$, and the KKT conditions leave two cases. Either $\lambda = 0$ and the least-squares $\hat{\theta}$ already fits inside the ball, or $\lambda > 0$ and the solution sits on the boundary, $\theta^T \theta = S$ (complementary slackness). Each S matches a λ : here $\lambda = 2$ goes with $S = 0.387$. The contours of $X^T X$ are tilted ellipses, so the touching point is not on the line from 0 to $\hat{\theta}_{\text{OLS}}$.

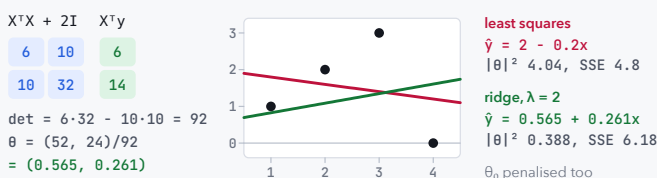
4 A zero gradient gives $(X^T X + \lambda I) \theta = X^T y$

Expand and differentiate, as for least squares:

$$\begin{aligned} \nabla_{\theta} J &= \nabla_{\theta} [y^T y - 2y^T X\theta + \theta^T X^T X\theta + \lambda \theta^T \theta] \\ &= -2X^T y + 2X^T X\theta + 2\lambda \theta = 0 \\ &\Rightarrow \hat{\theta}_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T y. \end{aligned}$$

Least squares was $\hat{\theta}_{\text{OLS}} = (X^T X)^{-1} X^T y$: the only change is λ on the diagonal. For large λ , $\hat{\theta} \approx X^T y / \lambda \rightarrow 0$. With the plain I the intercept θ_0 is penalised too.

5 Worked example: $\lambda = 2$ cuts $\|\theta\|^2$ tenfold

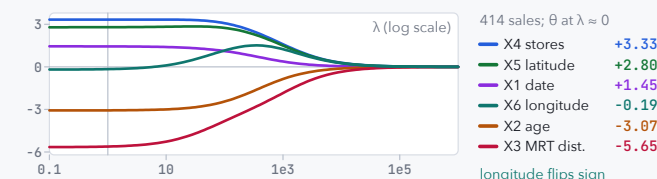


The outlier data of the linear regression I sheet: $X^T X = \begin{pmatrix} 4 & 10 \\ 10 & 30 \end{pmatrix}$, $X^T y = (6, 14)$, least squares $(2, -0.2)$. Ridge adds 2 on the diagonal. $\|\theta\|^2$ drops from 4.04 to 0.387, 9.6% of before, as on the slides. The training SSE rises from 4.8 to 6.18: smaller coefficients cost training fit, and only held-out data can say whether that pays.

λ shrinks θ and trades variance for bias

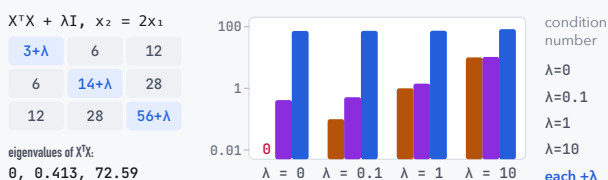
The coefficient path, why $X^T X + \lambda I$ is always invertible, the intercept, the gradient step, and λ from held-out data.

6 Coefficients shrink, never to exactly 0



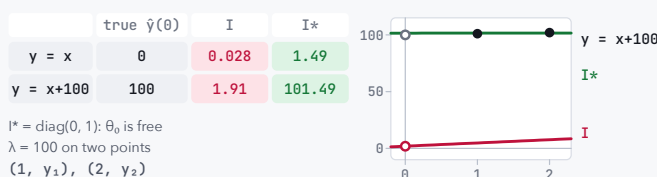
The slides' real-estate data: price per unit area on six standardised features. Every θ_j moves smoothly towards 0; at $\lambda = 10^6$ the smallest $|\theta_j|$ is still 0.0005. Longitude changes sign on the way, as correlated features shrink together. Ridge shrinks; the L1 penalty of lasso can zero coefficients out and so selects features (next sheet).

7 Adding λI makes the matrix invertible



The eigenvalues $d_j \geq 0$ of $X^T X$ become $d_j + \lambda \geq \lambda > 0$, so $X^T X + \lambda I$ is positive definite and $\hat{\theta}$ unique for every $\lambda > 0$, even with collinear columns or fewer rows than columns. The slides' X has $x_2 = 2x_1$ (rank 2; the linear regression II sheet). Its condition number $d_{\max}/d_{\min}$ falls from infinite to 727, 74 and 8.3; in the worked example $\lambda = 2$ takes it from 55.8 to 13.6.

8 Leave the intercept out of the penalty



The slides' test: $y = x$ and $y = x + 100$ at $x = 1, 2$, $\lambda = 100$, predict $\hat{y}(0)$. The plain I drags θ_0 towards 0: 1.91 instead of 100. With $I^* = \text{diag}(0, 1, \dots, 1)$, shifting y by 100 shifts the fit by exactly 100, so both errors are 1.49 ($\lambda = 100$ flattens the line to $\bar{y}$). Or centre x and y , use I , and add $\bar{y}$ back.

12 Pause and think

$\lambda \rightarrow \infty$? $\hat{\theta} \approx X^T y / \lambda \rightarrow 0$.

Does ridge drop features? No; lasso can.

More λ ? Bias up, variance down.

Why standardise? Units set the size of each θ_j .

Overfitting? Training error far below test error.

$X^T X$ singular? Fine for $\lambda > 0$.

minimise $\text{SSE} + \lambda \theta^T \theta \cdot \hat{\theta} = (X^T X + \lambda I)^{-1} X^T y \cdot \text{shrinks, never zeroes} \cdot \text{leave } \theta_0 \text{ free, standardise} \cdot \text{choose } \lambda \text{ by CV}$

Before: the matrix factorisation sheet; next: the lasso and coordinate descent sheet. Source: lecture slides "Ridge regression" and the ridge notebook with its 60-point dataset and real-estate table (nipunbatra.github.io/ml-teaching). The 20-point set, test points, noise redraws and CV shuffles are ours, seeded. Recomputed: the intercept example (I^* fits) and the gradient-step norms differ from the slides.

9 Gradient descent shrinks θ before each step



With $J = \frac{1}{2} \|y - X\theta\|^2 + \frac{\lambda}{2} \|\theta\|^2$ (same minimiser), $\nabla J = X^T X \theta - X^T y + \lambda \theta$ and

$$\theta^{(t+1)} = (1 - \alpha \lambda) \theta^{(t)} - \alpha (X^T X \theta^{(t)} - X^T y).$$

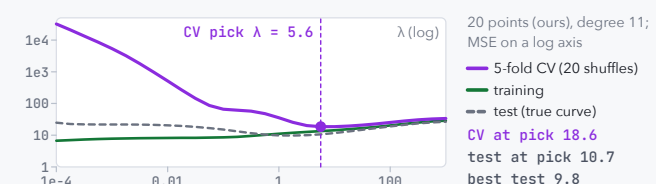
The factor $1 - \alpha \lambda < 1$ pulls θ towards 0 every step, which neural-network libraries call weight decay. The step size must stay below $2/(d_{\max} + \lambda) = 0.057$ here (the gradient descent sheet).

10 λ trades variance for bias



Redraw the noise 400 times, refit, and split the error at each λ as on the bias and variance sheet. At $\lambda = 10^{-4}$ the variance is 21.5 and the bias² 0.02; at $\lambda = 1000$ the bias² is 17.4 and the variance 0.46. Near $\lambda = 1.8$ the expected test MSE is 10.97, close to the noise floor of 9. Small λ overfits, large λ underfits.

11 Pick λ by cross-validation on a log grid



For each λ : fit on $k - 1$ folds, score the held-out fold, average over folds. Take the λ with the lowest mean, then refit on all the training data (the cross-validation sheet). Training error only rises with λ , so it cannot choose. Here CV picks 5.6, whose test MSE is 10.7 against a best of 9.8 on the grid.